No torus with ten convex quadrilateral faces

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Abstract

1 Introduction

2 Statement

We use the definitions of Grünbaum and Szilassi [1, p. 21].

- Faces are planar simple polygons with no collinear adjacent edges.
- The faces at a vertex form one circuit.
- Adjacent faces are not coplanar.
- The polyhedron is acoptic: every component of the intersection of two faces is a common vertex or a common edge.

Theorem 1. *No acoptic polyhedral torus in $\mathbb{R}^3$ has exactly ten faces that are all strictly convex quadrilaterals.*

Theorem 2. *Let $Q_i = (i, i+1, i+8, i+7)$, indices mod 10, be the faces of the map $\{4,4\}_{(3,1)}$. The map has no realisation in $\mathbb{R}^3$ with planar, strictly convex faces and non-coplanar adjacent faces, even if faces may intersect.*

Corollary 3. *Assume the eleven-face witness of [3] is correct, and assume [1, Prop. 3.1]. Then an acoptic torus with F strictly convex quadrilateral faces exists if and only if $F = 9$ or $F \geq 11$.*

3 Reduction to one map

Lemma 4. *In a torus as in Theorem 1, $V = 10$, $E = 20$ and every vertex has degree 4.*

Proof outline.

- From $4F = 2E$ and $V - E + F = 0$ we get $E = 20$ and $V = 10$.
- Two convex faces have a convex intersection. Being acoptic, it is empty, a common vertex or a common edge.
- At a vertex v of degree d, the points v, its d neighbours and its d opposite corners are pairwise distinct. A coincidence would make some segment vw a diagonal of one face lying in a second face, which the previous point forbids.

- Hence $2d+1 \le 10$, so $d \le 4$. The degree sum is $40 = 4V$, so every vertex has degree 4.

□

Lemma 5. *The face lattice is that of* $\{4,4\}_{(3,1)}$.

Proof outline.

- A degree-4 quadrangulation of the torus is $\mathbb{Z}^2/L$ with $[\mathbb{Z}^2 : L] = 10$.
- By Lemma 4, $L \cap \{-2,\dots,2\}^2 = \{0\}$.
- In Hermite normal form $L = \langle (a,0),(b,d)\rangle$ with $ad = 10$, this leaves only $L = \langle (10,0),(3,1)\rangle$ and its mirror image.
- Two independent exhaustive enumerations confirm this.

□

4 Sign conditions and the SAT certificate

Write $\chi(p_1,\dots,p_4) = \operatorname{sign}\det[1\ p_i]$. Every realisation satisfies the following.

(GP) For all a, b and all distinct $y_1,\dots,y_4$, the signs of $[aby_1y_2][aby_3y_4]$, $-[aby_1y_3][aby_2y_4]$ and $[aby_1y_4][aby_2y_3]$ are all zero or contain both signs (the three-term Plücker identity).

(PL) $\chi(Q_i) = 0$ for every face.

(AD) If faces F and G share an edge ab, c is another vertex of F, and x, y are the vertices of G off ab, then $\chi(a,b,c,x) = \chi(a,b,c,y) \neq 0$. Reason: G is strictly convex, and its plane meets plane(F) exactly in the line ab.

Lemma 6. *GP, PL and AD for* $\{4,4\}_{(3,1)}$ *are jointly unsatisfiable.*

- The CNF has 420 sign variables and 133 310 clauses.
- CaDiCaL proves it unsatisfiable. The LRAT proof (92 KB) is accepted by the CakeML-verified checker `cake_lpr` and by a second checker.
- A separately written encoder (no auxiliary variables) gives a CNF that is also unsatisfiable, with a verified LRAT proof.
- Both encoders are tested on genuine nine- and eleven-face tori: every clause holds for their exact chirotopes.
- Already the 8 faces left after deleting Q_0 and Q_3 give an unsatisfiable CNF, with verified LRAT proofs from both encoders. Inside it, a solver-reported group-minimal core uses 16 AD conditions and 37 GP relations; its minimality is not certificate-checked.

Lemmas 4–6 prove Theorem 1. Lemma 6 does not use non-intersection, which gives Theorem 2.

5 Data and verification

Code, certificates and a one-command checker: `%LEE:repository/DOIlink`.

- The SHA-256 hash of the core CNF is `9afbf70b30fd88a8c95839f163f25f1ee21c89f380faf986a1284b239938f4a2`.
- The check needs no human-readable proof of Lemma 6. Such a proof is not available.
- Residual literature risk: Simutis's 1977 thesis was not obtained.

6 Discussion

AI use

References

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